Prashant Maurya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.49
R B Children's Valley	Class XII (CBSE)	2023	89.80%
Sainik School Ghorakhal	Class X (CBSE)	2021	94.20%

WORK EXPERIENCE

10xSCALE AI - AI SOFTWARE ENGINEERING INTERN

Persistent Agent Memory & Semantic Retrieval

- Architected a **long term memory layer** using an async BaseStore CRUD abstraction with **3 interchangeable backends, Qdrant, Mem0, and in memory**, plus pluggable OpenAI and Google embeddings.
- Implemented **3 retrieval modes, preload, postload, and no retrieval**, with cross thread recall and LLM callable memory tools for **per user semantic memory across sessions**.
- Engineered **two tier memory deduplication** using metadata key matching and a **0.95 cosine similarity threshold** to prevent redundant memory persistence.
- Built **non blocking background writes** with a bounded **1,000 task queue**, pre read flushing, and indexed Qdrant metadata for consistent and filtered semantic retrieval.

Multimodal Agent I/O & Media Abstraction

- Extended the text only message architecture into a **Pydantic discriminated union supporting different content block types** with a unified MediaRef for URLs, provider files, base64 payloads, and internal media.
- Designed a pluggable media storage layer with **3 backends, local filesystem, in memory, and S3/GCS**, with automatic offloading of payloads larger than **50 KB**.
- Built a unified **MediaResolver** with model capability routing and transport fallback across **remote URLs and provider files.**

PROJECTS

Distributed Expense Settlement Engine

- Architected the **C++ backend** for a **scalable expense tracking platform**, engineering an **O(N log N) Max-Heap** and **Disjoint Set Union (DSU)** debt simplification algorithm to efficiently resolve simulated graphs of **100,000+ multi-party settlements**.
- Designed a **PostgreSQL double-entry ledger** utilizing **Optimistic Concurrency Control (OCC)** to eliminate **race conditions.**
- Integrated an **event-driven message queue** and **Redis cache architecture**, strictly enforcing **ACID properties** and guaranteeing **zero data loss** while achieving a load-tested throughput of **2,000+ TPS** at **sub-50ms latency**.

Inventory & Discount Management System

- Built a **feature-rich, console-based shopping cart** by applying **core data structures** to efficiently manage inventory.
- Utilized **C++ maps** for inventory management, **heaps** for **priority-based discount application**, and **file I/O** for **CSV data.**
- Delivered a **menu-driven user interface** for all cart operations, showcasing practical application of **data structures**.

Unsupervised Color Quantization for Image Compression

- Implemented **K-Means, Mini-Batch K-Means, and GMM** for RGB image compression using pixel-color cluster centers.
- Compared **hard vs. soft clustering** to evaluate trade-offs in compression efficiency, computational cost, and image quality.
- Conducted **quantitative** and **visual evaluation** in Python to compare reconstruction quality and performance.

SKILLS

Languages: C, C++, Python, SQL, Verilog

Agentic Toolkits: Langchain, Langraph, Fast APIs, RAG, Docker

Libraries: PyTorch, TensorFlow, NumPy, Pandas, Scikit-learn, Matplotlib

RELEVANT COURSES

Computer Science: Data Structures & Algorithms, OOPs, Operating Systems, DBMS, System Design

AI/ML: Foundation of Machine Learning, Deep Learning, Natural Language Processing

Mathematics: Probability and Random Variables, Introduction to Statistics, Linear Algebra, Transform Techniques, Calculus

SCHOLASTIC ACHIEVEMENTS

Qualified as a **Stage-II** Scholar in the National Talent Search Examination **(NTSE)**.

Ranked **874th among 500,000+ students in KVPY**, earning an **annual scholarship from the Indian Institute of Science (IISc).**

Secured a **Top 10 position** in the **Brand & Buzzword Hackathon** conducted by **Meltwater**, competing at the **PAN India level**.

POSITIONS OF RESPONSIBILITY

Teaching Assistant | Maths for Electrical Engineering (EE1010)

Guided students in core mathematical concepts, problem-solving, and evaluation for the course

Coordinator | Internship and Placement Cell | Office and Career Service

Pitched companies and coordinated internship and placement drives for smooth recruitment

Core | Rang de Manch(RDM) | Dramatics and Poetry Club

EXTRACURRICULAR

Represented **IIT Hyderabad** in **Mime** at **Inter IIT Cultural Meet 6.0**, securing **6th position** among 23 IITs

Completed a foundational **German** course with essential skills in communication, and basic grammar

Participated in **NSO Athletics** through regular training and involvement in athletic programs